

Section: Particles and Waves

Topic: Interference

The question involves the use of a double-slit experiment to investigate interference patterns.

(a)

State the type of wave behaviour shown

✓ The pattern shows **interference** — a property unique to waves where two or more waves overlap and combine.

(b)

State the change observed in the interference pattern when the distance between the slits is increased

✓ If slit separation increases:

- The **fringe spacing** decreases
- Fringes move closer together

Answer: **The fringes become closer together**

(c)

Calculate the distance between the slits

Use the formula for fringe spacing:

$$w = \frac{\lambda D}{d} \Rightarrow d = \frac{\lambda D}{w}$$

Given:

- $\lambda = 5.0 \times 10^{-7} \text{ m}$
- $D = 1.0 \text{ m}$
- $w = 2.5 \times 10^{-3} \text{ m}$

$$d = \frac{5.0 \times 10^{-7} \times 1.0}{2.5 \times 10^{-3}} = 2.0 \times 10^{-4} \text{ m}$$

Revision Tips

- Interference patterns form **only with coherent light** (same frequency and phase)
- Decreasing slit separation $d \rightarrow$ increases fringe spacing w
- Use:

$$w = \frac{\lambda D}{d}$$

where w = fringe spacing, λ = wavelength, D = distance to screen, d = slit separation

- Make sure all distances are in **metres**